BRNO UNIVERSITY OF TECHNOLOGY FACULTY OF ELECTRICAL ENGINEERING AND COMMUNICATION

Robotics and AI

MPC-MAP Assignment No. 5 - Report

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Task 3

I used the covariance matrix measured in the first 100 iterations for the KF measurement noise matrix Q. It defines how much I trust the GNSS measurement. I started with the given values for matrix R, which specifies how much I trust the prediction model. With this initial setup the estimation was noisy, so I needed to put more trust into the prediction model by decreasing R.

I found the point where the drift from the prediction became significant at R = $diag(1 \cdot 10^{-6},\ 1 \cdot 10^{-6},\ 1 \cdot 10^{-6})$. So finally, I set the parameters to the midpoint of the interval $(1 \cdot 10^{-6}; 0{,}01\ )$ and fine-tuned them to improve the result. The final values are $diag(8 \cdot 10^{-4},\ 8 \cdot 10^{-4},\ 7 \cdot 10^{-5}$ for a slightly noisier estimation but a smaller cross-track error, and $diag(3{,}8 \cdot 10^{-4},\ 3{,}8 \cdot 10^{-4},\ 5 \cdot 10^{-5})$ for a smooth estimation but with slightly less accurate trajectory tracking.

I also noticed that decreasing the maximum forward velocity of the robot helps, as the pose angle estimation improves due to more frequent GNSS updates. A successful ride is shown in Figure 1.

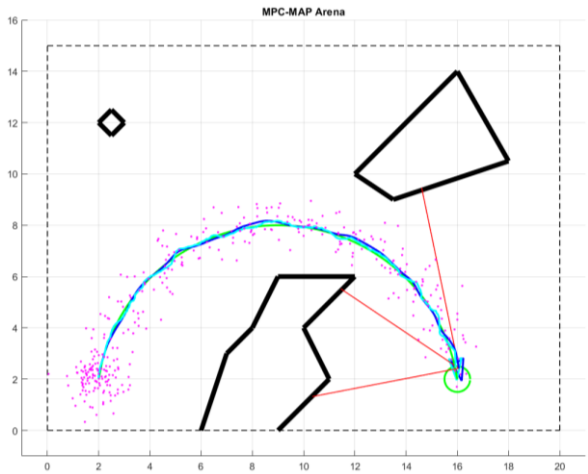

Fig. 1 – Successful ride with known initial pose

Task 4

For the initial position I am using the mean of the first 100 GNSS samples, and for the initial angle I set $\varphi = 0°$. For the initial covariance matrix $\sum$ I used the measured variances from the first 100 GNSS samples for x and y, and for φ I used π^2 to reflect the complete uncertainty in the initial orientation.

The problem I had was that at the beginning, when the estimated pose did not have the correct angle, the robot sometimes crashed into the wall (or the boundary). To address this, I used the trace of the covariance matrix $\sum$ to adjust the robot's forward speed, so when the uncertainty is high, the robot moves slowly.

The estimation from Task 3 is good enough, the real pose of the robot is recovered quickly, so I did not adjust the parameters of matrix R. A successful ride is shown in Figure 2.

Appendix

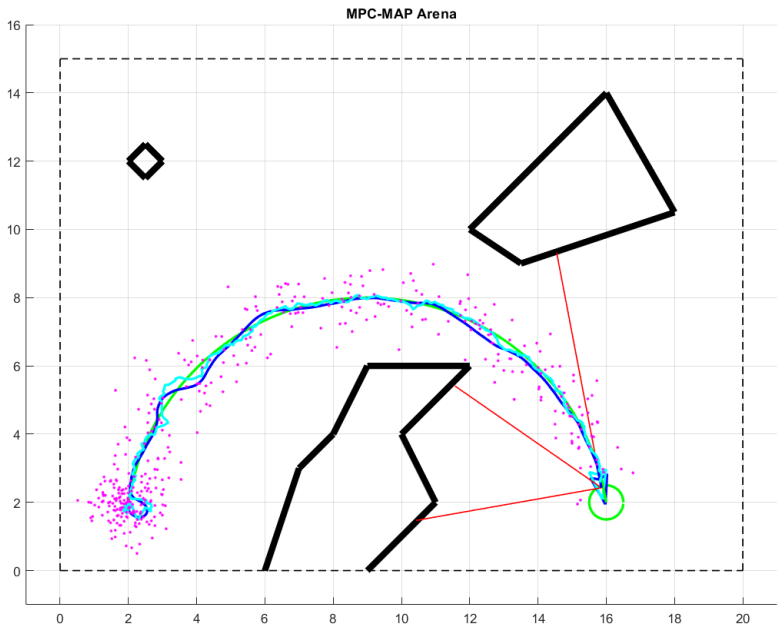

Fig. 2 – Successful ride with unknown initial pose of the robot